Tutorial – 02

Step 1: Understand and Define the Problem (Analyze)

Design a combinational logic circuit that operates an alarm in a car whenever driver or/and passenger seat/s are occupied.

The alarm must be activated in the following scenario/s.

- When the ignition is on and the driver or/and passenger seat are occupied and either driver seat belt or passenger seatbelt or both are unfastened.

Step 2: Organize and Describe the Data

This has 5 inputs

- DRIV - Driver is present (=1)
- PASS – Passenger is present (=1)
- IGN – Ignition switch on (=1)
- BELTD – Driver seatbelt Unfastened (=0)
- BELTP – Passenger seatbelt Unfastened (=0)

One output

- ALARM – LOW (Alarm is ON)

Step 3: Plan the Solution (Design the Algorithm)

3.1 Algorithm (in plain English)

- Monitor ignition status (IGN = 1)
- Check if the driver is seated and belt is not fastened. (DRIV = 1 | BELTD = 0)
- Check if the passenger is seated and belt is not fastened. (PASS = 1 | BELTP = 0)
- If either of the conditions are TRUE, and IGN = 1, the alarm activates.

- Otherwise, alarm remains HIGH. (off)

3.2 Truth Table

DRIV	BELTD	PASS	BELTP	IGN	ALARM
0	0	0	0	1	1
0	0	0	1	1	1
0	0	1	0	1	1
0	0	1	1	1	1
0	1	0	0	1	1
0	1	0	1	1	1
0	1	1	0	1	1
0	1	1	1	1	1
1	0	0	0	1	0
1	0	0	1	1	0
1	0	1	0	1	0
1	0	1	1	1	0
1	1	0	0	1	1
1	1	0	1	1	1
1	1	1	0	1	0
1	1	1	1	1	1

3.3 Boolean Expressions (SOP)

DRIV.$\overline{BELTD}.\overline{PASS}.\overline{BELTP}$.IGN

DRIV.$\overline{BELTD}.\overline{PASS}.BELTP$.IGN

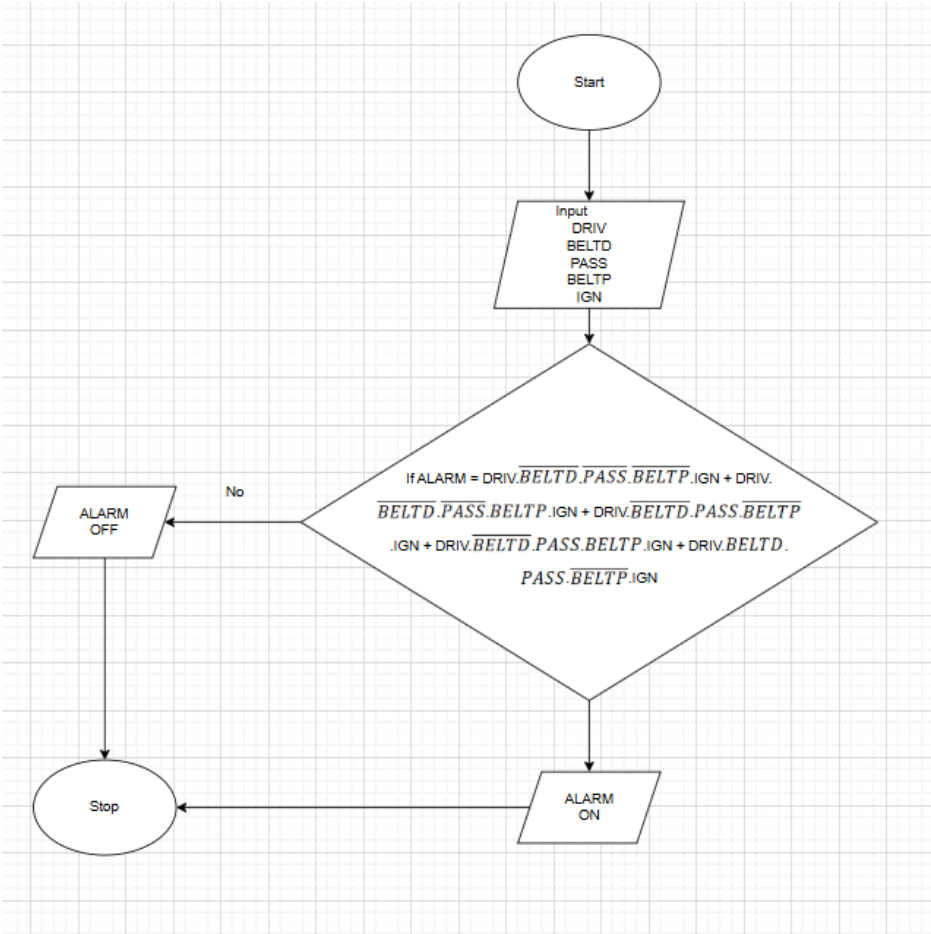
DRIV.$\overline{BELTD}.PASS.\overline{BELTP}$.IGN

DRIV.$\overline{BELTD}.PASS.BELTP$.IGN

DRIV.$BELTD.PASS.\overline{BELTP}$.IGN

ALARM = DRIV.$\overline{BELTD}.\overline{PASS}.\overline{BELTP}$.IGN + DRIV.$\overline{BELTD}.\overline{PASS}.BELTP$.IGN + DRIV.$\overline{BELTD}.PASS.\overline{BELTP}$.IGN + DRIV.$\overline{BELTD}.PASS.BELTP$.IGN + DRIV.$BELTD.PASS.\overline{BELTP}$.IGN

3.4 Flowchart



3.5 Pseudocode

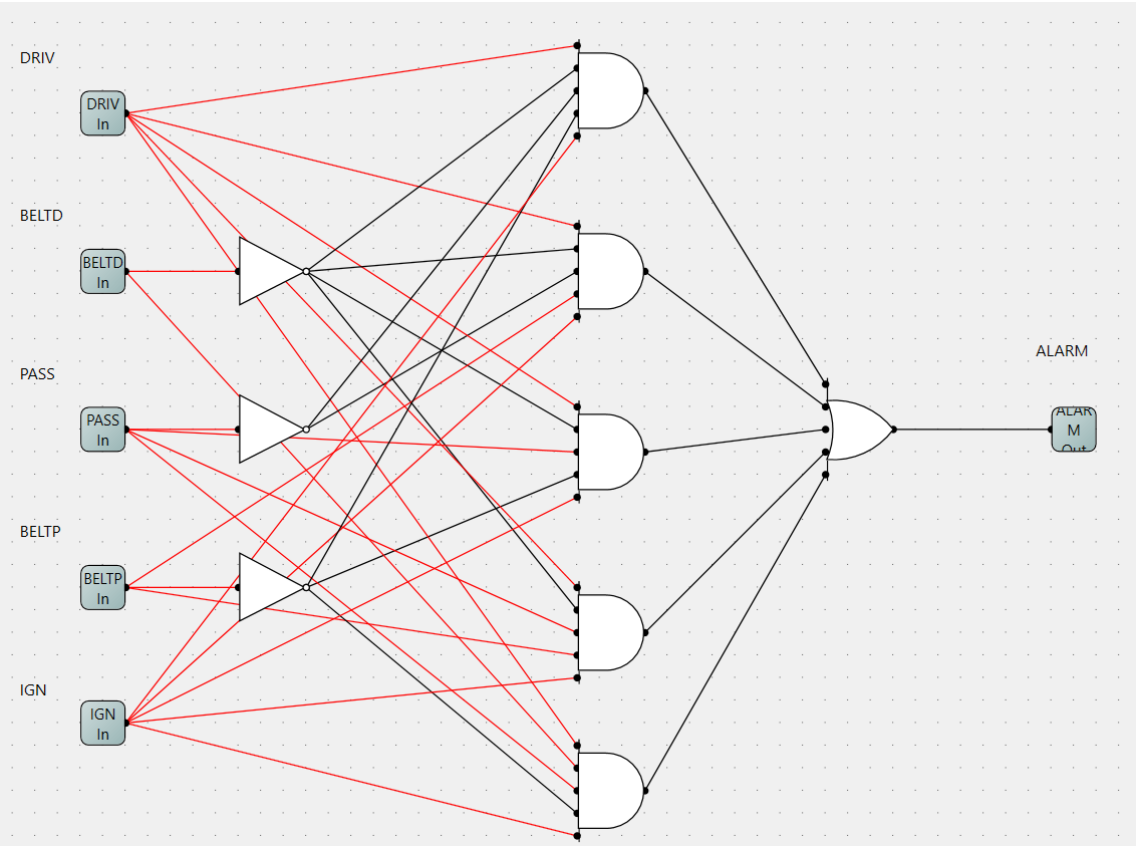
START

GET IGN, DRIV, BELTD, PASS, BELTP

IF DRIV.BELTD. PASS. BELTP.IGN + DRIV.BELTD. PASS.BELTPIGN + DRIV.BELTD. PASS. BELTP.IGN + DRIV.BELTD.PASS.BELTP.IGN + DRIV.BELTD. PASS.BELTP.IGN

THEN ALARM = 0 # ON

ELSE ALARM = 1 # OFF

END

Step 4: Test and Refine the Solution (Debug and Verify)